

DECO7230 – Prototype 2a Week 9 Test Plan

Tiny Worlds – Ladybird Anatomy Exploration

Target: ~5 min / participant

1. Test Goal

Check whether participants can independently understand the selection flow, discover the three interactive ladybird body parts, use the feedback (glow, cards, narration, animation and sound), and complete the exploration without excessive help. Collect one clear direction for Prototype 2b.

2. Main Assumptions to Validate

- A1 Participants understand how to select the available insect and recognise unavailable insects.
- A2 Participants can discover all 3 ladybird parts without being told their names/locations.
- A3 Blue glow + short card + narration make the interactions understandable.
- A4 The completion feedback makes it clear that the ladybird exploration is finished.
- A5 Participants can suggest what they want to do next, giving evidence for Prototype 2b.

3. 5-Minute Procedure

Time	Facilitator action	Record
0:00–0:30	Fit headset / confirm controls. Say only: “Explore the prototype naturally. I won’t guide you unless you get stuck.”	Start time
0:30–3:00	Participant uses Selection and explores Ladybird. Do NOT name the 3 body parts. If stuck for ~15 sec, give one small hint.	Ticks, hints, hesitation
3:00–4:00	Ask the 3 quick knowledge questions and the 6 rating items.	Short answers / ratings
4:00–5:00	Ask the 3 short Prototype 2b questions and one final comment.	Prototype 2b evidence

4. Facilitator Rules

- Do not explain where the antennae, elytra or wings are.
- Record help only when you actually intervene.
- Keep comments neutral.
- Use the participant sheet on page 2; one copy per tester.

5. Five XR Design Principles Check (after all sessions)

Principle	What to evaluate	1–5	Evidence / issue
Text Readability	Can cards and selection text be read quickly at viewing distance?	1 2 3 4 5	
Ergonomics	Are pointing, viewing height, and interaction duration comfortable?	1 2 3 4 5	
Brightness Changes	Is blue glow noticeable without being harsh?	1 2 3 4 5	
Simulator Sickness	Any discomfort, disorientation, or visual instability? Note simulator limits.	1 2 3 4 5	
Button Placement	Are Start / selection / Back controls easy to locate and activate?	1 2 3 4 5	

Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: __1__

Researcher records A and D. Participant answers B and C after the task.

A. Task Performance — Researcher observation

Measure	Record
Selected Ladybird successfully?	☐ Yes
Help needed?	☐ 1 hint
Understood a 'Coming Soon' insect?	☐ Yes
Antennae discovered?	☐ Yes
Elytra discovered?	☐ Yes
Flight wings discovered?	☐ Yes
Reached completion message?	☐ Yes

B. Quick Knowledge Check — Short answers

Question	Participant response	Correct?
What do the antennae help the ladybird do?	chemoreception	☐ Yes
What is the job of the elytra?	Protect the more delicate hind wings	☐ Yes
What are the flight wings used for?	Fly	☐ Yes

C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood what I was expected to do.	☐	☐	☐	☐	5
The blue glow made interactive parts easy to find.	☐	☐	☐	☐	5
The card + voice explanation was easy to understand.	☐	☐	☐	4	☐
The animations and sounds made my actions feel clear.	☐	☐	☐	☐	5
Overall, the prototype was easy to use.	☐	☐	☐	☐	5
The Ladybird exploration mission was easy to understand and complete.	☐	☐	☐	☐	5

D. Observation & Prototype 2b Evidence

Where did the participant hesitate?	Change perspective
What caused confusion?	3D vertigo
Which feedback helped most?	☐ Animation
What should you be able to do next?	☐ Feeding ☐ Habitat
What would you most like to learn about the ladybird next?	☐ What it eats ☐ Where it lives
What would make the next prototype feel more interactive?	☐ Feeding the ladybird ☐ Choosing/changing its habitat ☐ Guiding its movement ☐ Environmental response

Participant quote / other comment: “__I didn’t notice what the background voice is talking about__”

Principle	What to evaluate	1–5	Evidence / issue
Text Readability	Can cards and selection text be read quickly at viewing distance?	4	
Ergonomics	Are pointing, viewing height, and interaction duration comfortable?	5	
Brightness Changes	Is blue glow noticeable without being harsh?	5	
Simulator Sickness	Any discomfort, disorientation, or visual instability? Note simulator limits.	5	
Button Placement	Are Start / selection / Back controls easy to locate and activate?	5	

Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: __2__

Researcher records A and D. Participant answers B and C after the task.

A. Task Performance — Researcher observation

Measure	Record
Selected Ladybird successfully?	☐ Yes
Help needed?	☐ 1 hint
Understood a 'Coming Soon' insect?	☐ Yes
Antennae discovered?	☐ Yes
Elytra discovered?	☐ Yes
Flight wings discovered?	☐ Yes
Reached completion message?	☐ Yes

B. Quick Knowledge Check — Short answers

Question	Participant response	Correct?
What do the antennae help the ladybird do?	Touch and smell	☐ Yes
What is the job of the elytra?	Protect the flight wings	☐ Yes
What are the flight wings used for?	Fly	☐ Yes

C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood what I was expected to do.	☐	☐	☐	4	☐
The blue glow made interactive parts easy to find.	☐	☐	☐	☐	5
The card + voice explanation was easy to understand.	☐	☐	3	☐	☐
The animations and sounds made my actions feel clear.	☐	☐	3	☐	☐
Overall, the prototype was easy to use.	☐	☐	☐	4	☐
The Ladybird exploration mission was easy to understand and complete.	☐	☐	☐	☐	5

D. Observation & Prototype 2b Evidence

Where did the participant hesitate?	Didn't notice the voice guidance.
What caused confusion?	Don't know what to do next
Which feedback helped most?	☐ Animation
What should you be able to do next?	☐ Movement/behaviour
What would you most like to learn about the ladybird next?	☐ What it eats ☐ How it behaves/
What would make the next prototype feel more interactive?	☐ Feeding the ladybird ☐ Guiding its movement

Participant quote / other comment: “__ I feel sick when the ladybird come so close to me __”

Principle	What to evaluate	1–5	Evidence / issue
Text Readability	Can cards and selection text be read quickly at viewing distance?	4	
Ergonomics	Are pointing, viewing height, and interaction duration comfortable?	4	
Brightness Changes	Is blue glow noticeable without being harsh?	4	
Simulator Sickness	Any discomfort, disorientation, or visual instability? Note simulator limits.	3	Too close to the bug

Button Placement	Are Start / selection / Back controls easy to locate and activate?	4	
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Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: __3 s__

Researcher records A and D. Participant answers B and C after the task.

A. Task Performance — Researcher observation

Measure	Record
Selected Ladybird successfully?	☐ Yes
Help needed?	☐ 1 hint
Understood a 'Coming Soon' insect?	☐ Yes
Antennae discovered?	☐ Yes
Elytra discovered?	☐ Yes
Flight wings discovered?	☐ Yes
Reached completion message?	☐ Yes

B. Quick Knowledge Check — Short answers

Question	Participant response	Correct?
What do the antennae help the ladybird do?	Touch and smell	☐ Yes
What is the job of the elytra?	Protect the wings	☐ Yes
What are the flight wings used for?	Use to fly	☐ Yes

C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood what I was expected to do.	☐	☐	☐	4	☐
The blue glow made interactive parts easy to find.	☐	☐	☐	4	☐
The card + voice explanation was easy to understand.	☐	☐	☐	☐	5
The animations and sounds made my actions feel clear.	☐	☐	☐	☐	5
Overall, the prototype was easy to use.	☐	☐	☐	☐	5
The Ladybird exploration mission was easy to understand and complete.	☐	☐	3	☐	☐

D. Observation & Prototype 2b Evidence

Where did the participant hesitate?	It's hard to find the wings can interact with
What caused confusion?	No
Which feedback helped most?	☐ Glow ☐ Animation
What should you be able to do next?	☐ Feeding ☐ Other: __ Control movement like make the ladybird fly.
What would you most like to learn about the ladybird next?	☐ Where it lives
What would make the next prototype feel more interactive?	☐ Feeding the ladybird ☐ Guiding its movement

Participant quote / other comment: “__ It would be great if can feed the insect __”

Principle	What to evaluate	1–5	Evidence / issue
Text Readability	Can cards and selection text be read quickly at viewing distance?	4	
Ergonomics	Are pointing, viewing height, and interaction duration comfortable?	5	
Brightness Changes	Is blue glow noticeable without being harsh?	5	yes

Simulator Sickness	Any discomfort, disorientation, or visual instability? Note simulator limits.	5	Not sick
Button Placement	Are Start / selection / Back controls easy to locate and activate?	5	

Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: __4__

Researcher records A and D. Participant answers B and C after the task.

A. Task Performance — Researcher observation

Measure	Record
Selected Ladybird successfully?	☐ Yes
Help needed?	☐ None
Understood a 'Coming Soon' insect?	☐ Yes
Antennae discovered?	☐ Yes
Elytra discovered?	☐ Yes
Flight wings discovered?	☐ Yes
Reached completion message?	☐ Yes

B. Quick Knowledge Check — Short answers

Question	Participant response	Correct?
What do the antennae help the ladybird do?	sense the smell	☐ Yes
What is the job of the elytra?	Protect the wings	☐ Yes
What are the flight wings used for?	Fly	☐ Yes

C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood what I was expected to do.	☐	☐	☐	☐	5
The blue glow made interactive parts easy to find.	☐	☐	☐	☐	5
The card + voice explanation was easy to understand.	☐	☐	☐	☐	5
The animations and sounds made my actions feel clear.	☐	☐	☐	☐	5
Overall, the prototype was easy to use.	☐	☐	☐	4	☐
The Ladybird exploration mission was easy to understand and complete.	☐	☐	☐	☐	5

D. Observation & Prototype 2b Evidence

Where did the participant hesitate?	No hesitate
What caused confusion?	The function part of the trigger need to change
Which feedback helped most?	☐ Glow ☐ Voice ☐ Sound
What should you be able to do next?	☐ Feeding ☐ Habitat
What would you most like to learn about the ladybird next?	☐ What it eats ☐ Where it lives
What would make the next prototype feel more interactive?	☐ Feeding the ladybird ☐ Choosing/changing its habitat

Participant quote / other comment: “__It could be very interesting if I can feed the lady bird or ride the ladybird__”

Principle	What to evaluate	1–5	Evidence / issue
Text Readability	Can cards and selection text be read quickly at viewing distance?	5	
Ergonomics	Are pointing, viewing height, and interaction duration comfortable?	5	

Brightness Changes	Is blue glow noticeable without being harsh?	5	
Simulator Sickness	Any discomfort, disorientation, or visual instability? Note simulator limits.	4	
Button Placement	Are Start / selection / Back controls easy to locate and activate?	4	

Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: __5__

Researcher records A and D. Participant answers B and C after the task.

A. Task Performance — Researcher observation

Measure	Record
Selected Ladybird successfully?	☐ Yes
Help needed?	☐ None
Understood a 'Coming Soon' insect?	☐ Yes
Antennae discovered?	☐ Yes
Elytra discovered?	☐ Yes
Flight wings discovered?	☐ Yes
Reached completion message?	☐ Yes

B. Quick Knowledge Check — Short answers

Question	Participant response	Correct?
What do the antennae help the ladybird do?	Sense touch and smell	☐ Yes
What is the job of the elytra?	Protect wings	☐ Yes
What are the flight wings used for?	Fly	☐ Yes

C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood what I was expected to do.	☐	☐	☐	☐	5
The blue glow made interactive parts easy to find.	☐	☐	☐	☐	5
The card + voice explanation was easy to understand.	☐	☐	☐	☐	5
The animations and sounds made my actions feel clear.	☐	☐	☐	☐	5
Overall, the prototype was easy to use.	☐	☐	☐	☐	5
The Ladybird exploration mission was easy to understand and complete.	☐	☐	☐	4	☐

D. Observation & Prototype 2b Evidence

Where did the participant hesitate?	Did find any part hesitate.
What caused confusion?	The design of the buttons on the trigger is a bit confuse.
Which feedback helped most?	☐ Glow ☐ Animation ☐ Voice
What should you be able to do next?	☐ Feeding ☐ Habitat
What would you most like to learn about the ladybird next?	☐ What it eats ☐ Where it lives
What would make the next prototype feel more interactive?	☐ Feeding the ladybird ☐ Choosing/changing its habitat ☐ Guiding its movement

Principle	What to evaluate	1–5	Evidence / issue
Text Readability	Can cards and selection text be read quickly at viewing distance?	5	

Ergonomics	Are pointing, viewing height, and interaction duration comfortable?	5	
Brightness Changes	Is blue glow noticeable without being harsh?	4	The wing is hard to notice
Simulator Sickness	Any discomfort, disorientation, or visual instability? Note simulator limits.	5	
Button Placement	Are Start / selection / Back controls easy to locate and activate?	5	